Ch. 7 — P01 (A) M/M/1 Formulas

Problem Bank | Chapter 7 | Type: A | Difficulty:

Chapter: 7 Type: A Difficulty:

A single-server service facility operates as an M/M/1 queue with arrival rate $\lambda = 6$ customers/hour and service rate $\mu = 8$ customers/hour.

Tasks

1. Compute the traffic intensity $\rho = \lambda/\mu$.
2. Compute the mean number of customers in the system (L) and in the queue (L_q).
3. Compute the mean time in the system (W) and in the queue (W_q). Express your answers in minutes.
4. Compute the probability that a customer must wait (i.e., $P(\text{wait} > 0) = \rho$).
5. Compute $P(N \geq 3)$: the probability that three or more customers are in the system.
6. What arrival rate λ^* would result in $W_q = 5$ minutes, given the same μ?

All results should be expressed to three significant figures.